

NETWORKS AND COMPLEXITY

Solution 14-4

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 14.4: Two abstract two-dimensional system [Lvl. 2]

Compute the steady states of the following systems and analyze their stability:

a) System 1

$$\begin{aligned}\dot{x} &= y^5 - x^2 \\ \dot{y} &= 1 - x\end{aligned}$$

Solution

From the equation for $\dot{y}$ we obtain the condition

$$0 = 1 - x \tag{1}$$

and hence

$$x^* = 1 \tag{2}$$

Substituting this into the condition from the equation for $\dot{x}$ yields

$$0 = y^5 - 1 \tag{3}$$

and hence

$$y = \sqrt[5]{1} = 1$$

We compute the Jacobian elements

$$J_{11} = \left. \frac{\partial \dot{x}}{\partial x} \right|_* = \left. \frac{\partial}{\partial x} (y^5 - x^2) \right|_* = -2x^* = -2 \tag{4}$$

$$J_{12} = \left. \frac{\partial \dot{x}}{\partial y} \right|_* = \left. \frac{\partial}{\partial y} (y^5 - x^2) \right|_* = 5y^{*4} = 5 \tag{5}$$

$$J_{21} = \left. \frac{\partial \dot{y}}{\partial x} \right|_* = \left. \frac{\partial}{\partial x} (1 - x) \right|_* = -1 \tag{6}$$

$$J_{22} = \left. \frac{\partial \dot{y}}{\partial y} \right|_* = \left. \frac{\partial}{\partial y} (1 - x) \right|_* = 0 \tag{7}$$

$$\tag{8}$$

and hence

$$\mathbf{J} = \begin{pmatrix} -2 & 5 \\ -1 & 0 \end{pmatrix} \tag{9}$$

We start from the eigenvector equation

$$\begin{pmatrix} -2 & 5 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \lambda \begin{pmatrix} A \\ B \end{pmatrix} \tag{10}$$

where A and B are placeholders for the eigenvector elements. Because we can rescale eigenvectors we can set $A = 1$ (unless A needs to be 0). Reading the two rows of the equation system separately gives us

$$-2 + 5B = \lambda \quad (11)$$

$$-1 = \lambda B \quad (12)$$

The second row tells us $B = -1/\lambda$, and substituting this into the first row gives us

$$-2 - \frac{5}{\lambda} = \lambda \quad (13)$$

Multiplying this equation by λ and bringing all terms over to the right side gives us the characteristic polynomial

$$0 = \lambda^2 + 2\lambda + 5 \quad (14)$$

which we now solve for λ .

$$\begin{aligned} 0 &= \lambda^2 + 2\lambda + 5 \\ 0 &= (\lambda + 1)^2 + 4 \\ -4 &= (\lambda + 1)^2 \\ \lambda &= -1 \pm \sqrt{-4} = -1 \pm 2\sqrt{-1} \end{aligned}$$

which we can also write as

$$\lambda_{1,2} = -1 \pm 2i \quad (15)$$

where i is the imaginary number. If we also want the eigenvectors we could now find B by using $B = -1/\lambda$, but dividing by complex numbers is tedious, so let's rather use the top row from the eigenvector equation ($-2 + 5B = \lambda$) which we can write as

$$B = \frac{2}{5} + \frac{1}{5}\lambda = \frac{1}{5} \pm \frac{2}{5}i \quad (16)$$

Because fractions in vectors don't look nice let's scale the whole eigenvectors by a factor of 5. The result are the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 5 \\ 1 + 2i \end{pmatrix} \quad \mathbf{v}_2 = \begin{pmatrix} 5 \\ 1 - 2i \end{pmatrix} \quad (17)$$

To compute the real part we discard the imaginary part of the complex eigenvalues

$$\text{Re}(\lambda_{1,2}) = \text{Re}(-1 \pm \sqrt{-4}) = -1 \quad (18)$$

So both eigenvalues have the real part -1. Because all real parts are negative the steady state is stable.

b) System 2

$$\dot{x} = x - y + 2 \quad (19)$$

$$\dot{y} = x - 3y + 8 \quad (20)$$

Solution

From the differential equation for x we get the condition

$$0 = x - y + 2 \quad (21)$$

and hence

$$y = x + 2 \quad (22)$$

The equation for the change of y gives us the condition

$$0 = x - 3y + 8 = x - 3(x + 2) + 8 = x - 3x - 6 + 8 = -2x + 2 \quad (23)$$

and hence

$$2x = 2 \quad (24)$$

Therefore the steady state is $x^* = 1$ and $y^* = 3$.

To find the Jacobian we compute the derivatives

$$J_{11} = \left. \frac{\partial \dot{x}}{\partial x} \right|_* = \left. \frac{\partial}{\partial x} x - y + 2 \right|_* = 1 \quad (25)$$

$$J_{12} = \left. \frac{\partial \dot{x}}{\partial y} \right|_* = \left. \frac{\partial}{\partial y} x - y + 2 \right|_* = -1 \quad (26)$$

$$J_{21} = \left. \frac{\partial \dot{y}}{\partial x} \right|_* = \left. \frac{\partial}{\partial x} x - 3y + 8 \right|_* = 1 \quad (27)$$

$$J_{22} = \left. \frac{\partial \dot{y}}{\partial y} \right|_* = \left. \frac{\partial}{\partial y} x - 3y + 8 \right|_* = -3 \quad (28)$$

Hence the Jacobian is

$$\mathbf{J} = \begin{pmatrix} 1 & -1 \\ 1 & -3 \end{pmatrix}. \quad (29)$$

We compute the characteristic polynomial

$$\left| \begin{pmatrix} 1 - \lambda & -1 \\ 1 & -3 - \lambda \end{pmatrix} \right| = \lambda^2 + 2\lambda - 2 \quad (30)$$

We solve for the eigenvalues

$$0 = \lambda^2 + 2\lambda - 2 \quad (31)$$

$$0 = (\lambda^2 + 2\lambda + 1) - 3 \quad (32)$$

$$3 = (\lambda + 1)^2 \quad (33)$$

$$\pm\sqrt{3} = \lambda + 1 \quad (34)$$

$$(35)$$

Therefore the eigenvalues are

$$\lambda_{1,2} = -1 \pm \sqrt{3} \quad (36)$$

Since $\sqrt{3}$ is a real number the real part of the eigenvalue is

$$\text{Re}(\lambda_1) = \lambda_1 = -1 + \sqrt{3} \quad (37)$$

Because $\sqrt{3} > 1$ this eigenvalue is positive which reveals that the steady state is unstable.